

William Lam Journal

Date: 15/03/2026

Goal: Understand Task

Description:

I began work on the task for the first time, creating the journal as well as a folder to hold my work. I read through the notification and task description and understand that I am to improve on the PWA given so that it is user-friendly, secure and efficient. Additionally I am to create a report detailing the changes and justifying them. I did not spend a lot of time this week on the task as I am dealing with other assessments currently.

Date: 20/03/2026

Goal: Begin basic additions

Description:

I started by cleaning up the original Learner Hours code, specifically the calling of images as they had different names to the files. In the [sw.js](#) file, there is a line referring to 'wheel.svg' as a constant image (Line 10). However, it is not given in the original task files. I will either delete it or ask for further clarification from teachers. There is inline styling in the 'index.html' file in line 20, '<div style="margin-left:30px;">'. I have flagged this and will come back to it in the future, with a decision to either move it to the css file or keep it. I also turned on the service worker which was commented out.

I created a backup of my files so I can revert back in the future and created a VSCode workspace so I can easily open up on both my laptop and home computer.

I started on part A. I added the text field to enter a description and got it working. I also added a cancel button that clears the session contents.

I added the 'edit' and 'delete' buttons. Delete button has functionality, but I am struggling with the edit functionality.

I began to add the search/filter function.

Date: 21/03/2026

Goal: Complete Editing and Filter functions

Description:

The editing function was completed. A global boolean variable needed to be declared (named 'editing' in mine) that determines whether or not the session is being edited. When

the button is clicked, the old session data is to be restored and then saved into an object called 'editingSession'. Once the new session is submitted, the submit listener checks the 'editing variable'. If true, it deletes the old session using data from editingSession. It then stores the new session as per the rest of the method call.

The search/filter function was completed. Works by creating an empty object called searchCriteria. Depending on the amount of input given by the user (date, start/end time, desc), it filters all previous sessions using the checkMatch function, with exact matches for date and timings, with descriptions only requiring a partial match. It then calls the renderPastSessions function with the new filtered sessions to render them.

A clear all function was added, which deletes all the sessions. Once the clear all button is pressed, all sessions are cleared. A confirmation popup was added.

Date: 24/03/2026

Goal: Fix up Cancel and Filters

Description:

Started by adding an event listener for the 'Cancel' Button, as there was a bug where the submit button did not turn its text content back to "Add Session" while editing a session and the cancel button was clicked. Added a button to the filters, which toggles between newest and oldest entries.

Date: 26/03/2026

Goal: rework Filter

Description:

Reworked the filter function. Instead of toggling a button, users will now be able to use a dropdown menu to choose from a variety of filters and choose whether to use ascending or descending order. Functionality has not been added yet.

Date: 27/03/2026

Goal: Continue and finish Filters

Description:

The aim today is to add functionality to the filters. I finished all the filters (Date, Start-Time, End-Time). I decided not to filter by description as I feel it would be redundant as the user can merely search by it. I removed the inline styling from the HTML into the css file. I added a total hours part which calculates the total number of hours.

Date: 28/03/2026

Goal: Continue Hours function

Description:

I created a session element that displays the hours per session. Removed the wheel.svg from the [sw.js](#) file (redundant). Did not have a lot of time this session.

30/03/2026

Goal: work heavily through Part B to E

Description:

Made a login form. Did parts B to E to the best of my ability. Programmed all CSS, setting colours and sizing buttons, as well as making it compatible with different browsers and screen sizes.. Struggled on some parts, including secure password policies. Not entirely sure if I am getting it correctly. Made the PWA usable for different screen sizes and different browsers. I implemented lazy loading on the list of past sessions, as a large number of them rendering at once can lag. Added DOMpurify which prevents XSS. The PWA has a maximum of 1 user (artificial limit set) for simplicity. Considered using terser online to compress JS files for faster loading.

31/03/2026

Goal: Finish Part E and complete Project Report, Submit

Description:

Used hashing to make password secure. Fixed various bugs relating to alerts. Began project report. I used minifiers on my files. Rewrote a bit of [sw.js](#) for caching strategy.

The project report was finished very early in the morning. It took a very long time and was challenging to complete.

All documents were zipped and submitted.